TurtleBot3 SLAM

Contents

TurtleBot3 SLAM

Exported from Confluence · https://pinkwink.atlassian.net

[SBC] 로봇 원격 접속하기

```
mkh@mkh:~$ ssh ubuntu@192.168.4.1
ubuntu@192.168.4.1's password:
Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 5.15.0-1059-raspi aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Aug  7 06:21:14 AM UTC 2024

  System load:  0.81               Temperature:           48.7 C
  Usage of /:   23.7% of 28.93GB   Processes:             148
  Memory usage: 6%                 Users logged in:       0
  Swap usage:   0%                 IPv4 address for wlan0: 192.168.0.110

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.
```

[SBC] 터틀봇 모델 설정되어 있는지 확인

Code

```
echo $TURTLEBOT3_MODEL
```

```
ubuntu@ubuntu:~/opencr_update$ echo $TURTLEBOT3_MODEL
burger
ubuntu@ubuntu:~/opencr_update$ 
```

[SBC] 로봇 실행 하기

Code

```
ros2 launch turtlebot3_bringup robot.launch.py
```

```
ubuntu@ubuntu:~$ ros2 launch turtlebot3_bringup robot.launch.py
[INFO] [launch]: All log files can be found below /home/ubuntu/.ros/log/2024-08-
[INFO] [launch]: Default logging verbosity is set to INFO
urdf_file_name : turtlebot3_waffle_pi.urdf
[INFO] [robot_state_publisher-1]: process started with pid [1491]
[INFO] [ld08_driver-2]: process started with pid [1493]
[INFO] [turtlebot3_ros-3]: process started with pid [1495]
[turtlebot3_ros-3] [INFO] [1723011762.098915685] [turtlebot3_node]: Init TurtleB
[turtlebot3_ros-3] [INFO] [1723011762.099445848] [turtlebot3_node]: Init Dynamix
[turtlebot3_ros-3] [INFO] [1723011762.102022627] [DynamixelSDKWrapper]: Succeede
[turtlebot3_ros-3] [INFO] [1723011762.111766581] [DynamixelSDKWrapper]: Succeede
[robot_state_publisher-1] [INFO] [1723011762.129627480] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.129939182] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.129997570] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130028255] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130054422] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130080866] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130106625] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130131532] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130157939] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130182976] [robot_state_publisher]:
[robot_state_publisher-1] [INFO] [1723011762.130208013] [robot_state_publisher]:
```

[PC] 새 터미널 열어서 PC에서도 로봇 모델 확인하기

Code

```
echo $TURTLEBOT3_MODEL
```

```
mkh@mkh:~$ echo $TURTLEBOT3_MODEL
burger
mkh@mkh:~$
```

[PC] 로봇과 잘 연결 되었는지 토픽 리스트 확인

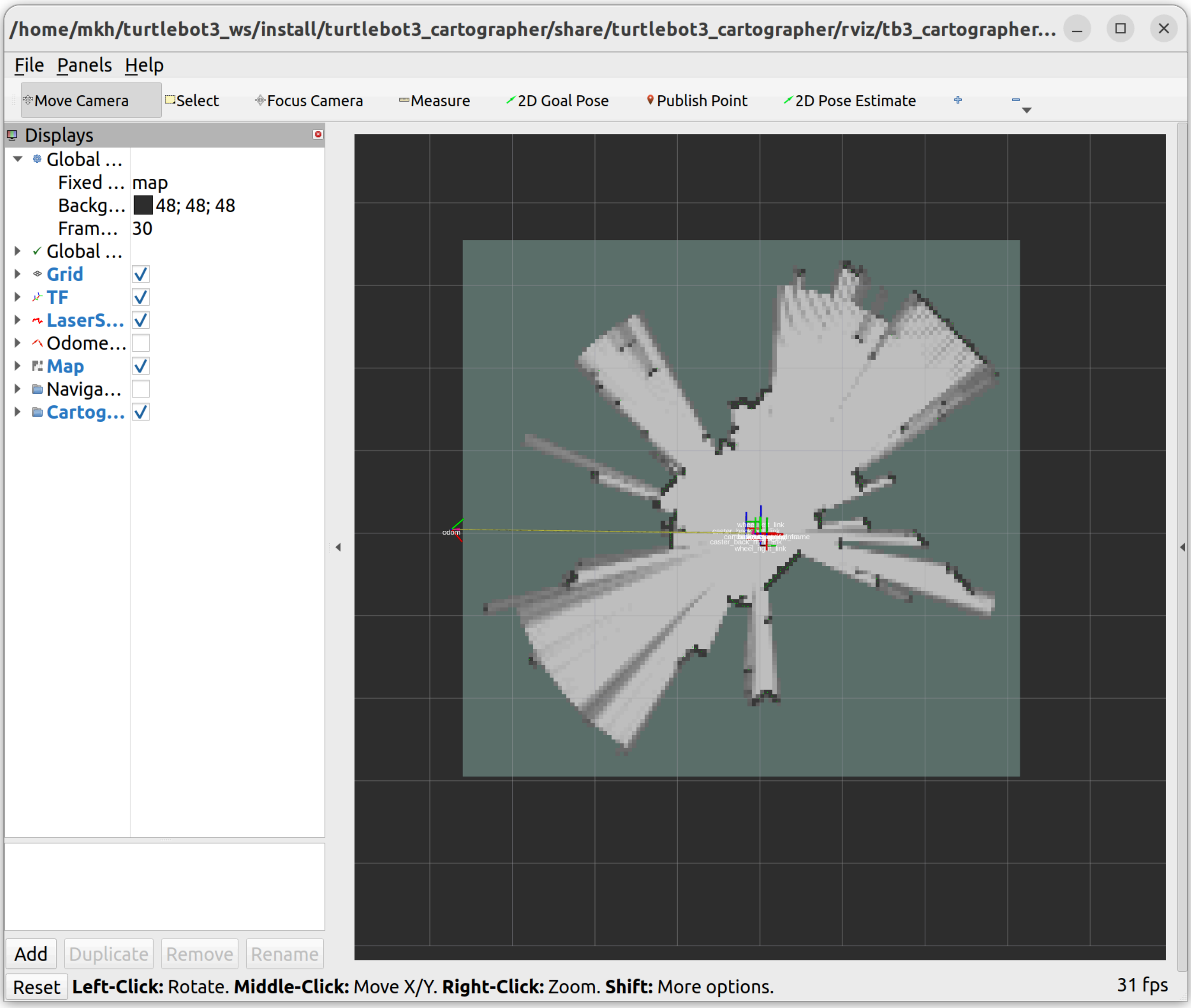
```
mkh@mkh:~$ ros2 topic list
/battery_state
/cmd_vel
/imu
/joint_states
/magnetic_field
/odom
/parameter_events
/robot_description
/rosout
/scan
/sensor_state
/tf
/tf_static
mkh@mkh:~$ 
```

[PC] SLAM 실행하기

Code

```
ros2 launch turtlebot3_cartographer cartographer.launch.py
```

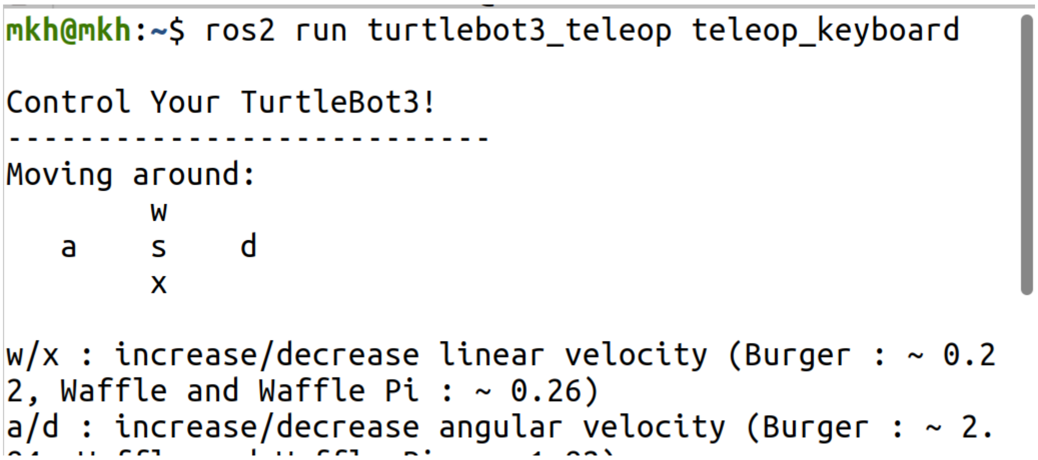
```
mkh@mkh:~$ ros2 launch turtlebot3_cartographer cartographer.launch.py
[INFO] [launch]: All log files can be found below /home/mkh/.ros/log/2024-08-07-15-37-10-708112-mkh-33102
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [cartographer_node-1]: process started with pid [33103]
[INFO] [cartographer_occupancy_grid_node-2]: process started with pid [33105]
[INFO] [rviz2-3]: process started with pid [33107]
[cartographer_node-1] [INFO] [1723012630.785384796] [cartographer logger]: I0807 15:37:10.000000 33103 configuration_file_resolver.cc:41] Found '/home/mkh/turtlebot3_ws/install/turtlebot3_cartographer/share/turtlebot3_cartographer/config/turtlebot3_lds_2d.lua' for 'turtlebot3_lds_2d.lua'.
[cartographer_node-1] [INFO] [1723012630.785565542] [cartographer logger]: I0807 15:37:10.000000 33103 configuration_file_resolver.cc:41] Found '/opt/ros/humble/share/cartographer/configuration_files/map_builder.lua' for 'map_builder.lua'.
```



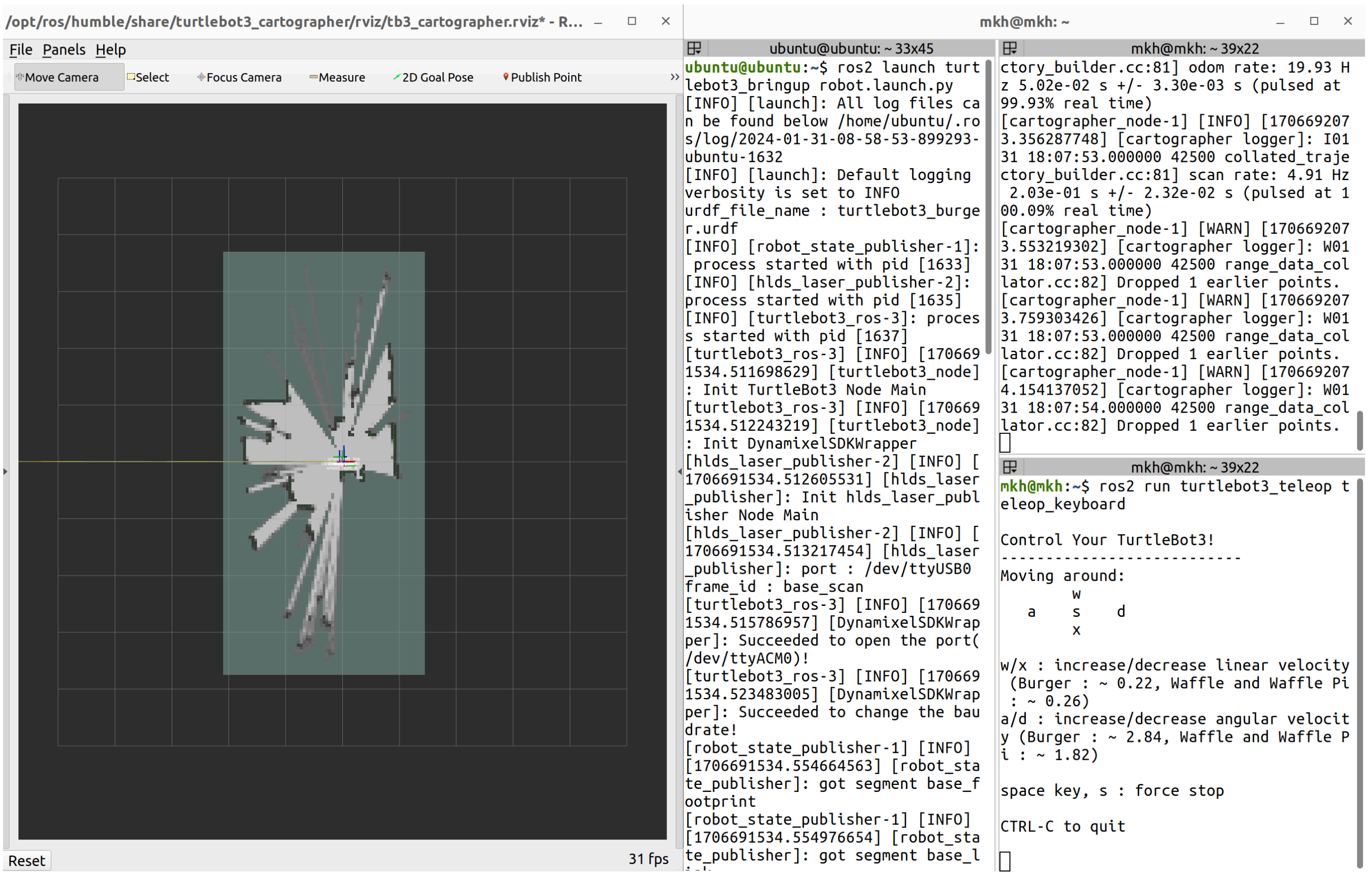
[PC] 새 터미널 하나 더 열어서 터틀봇3 키보드 제어 노드 실행하기

Code

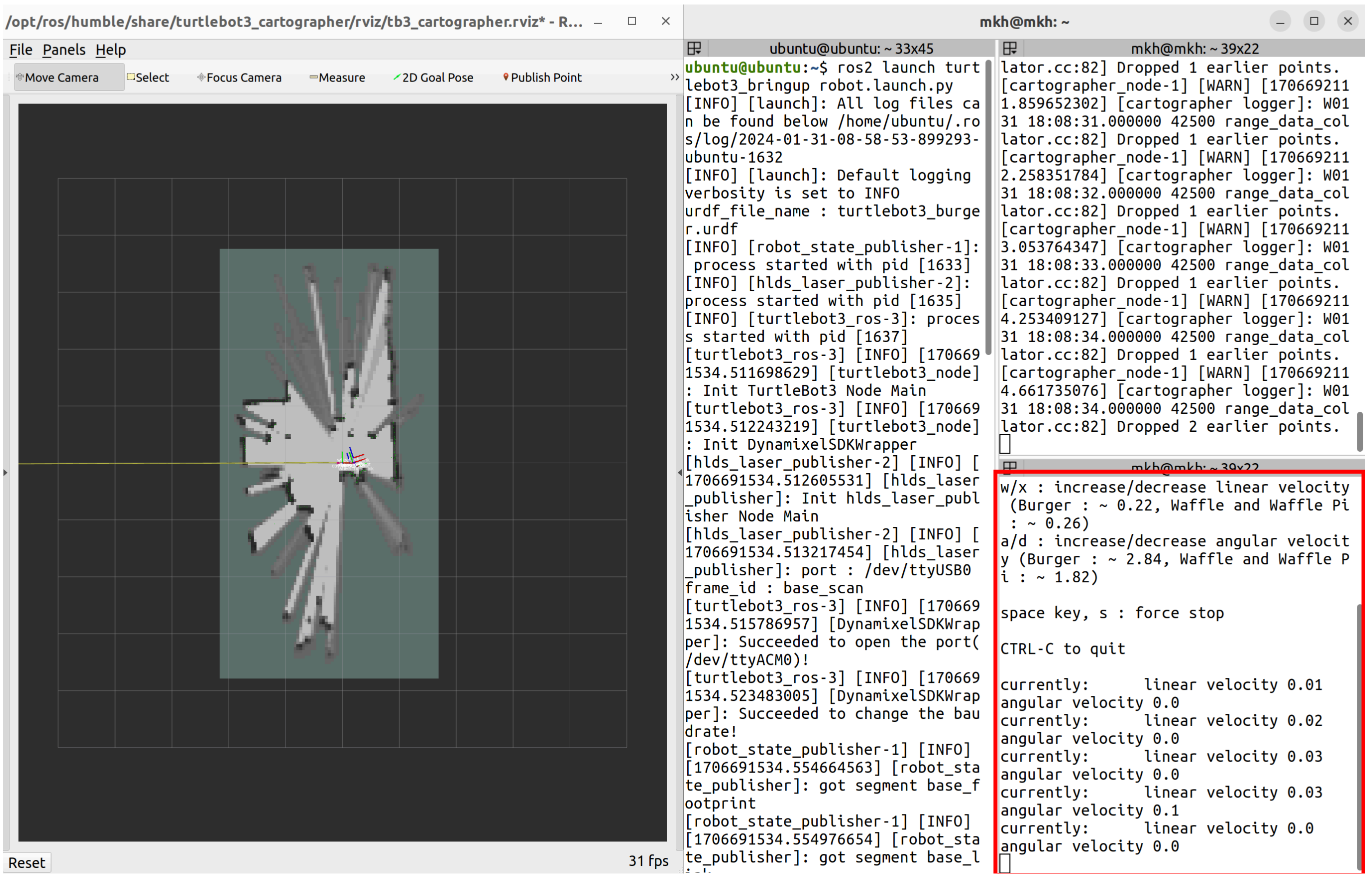
```
ros2 run turtlebot3_teleop teleop_keyboard
```



이후 화면 배치를 잘 해서



터틀봇3 키보드 제어 터미널에 w, a, s, d, x를 입력에 로봇을 제어하면서 맵 만들기


/opt/ros/humble/share/turtlebot3_cartographer/rviz/tb3_cartographer.rviz* - R...
File Panels Help
Move Camera
Select
Focus Camera
Measure
2D Goal Pose
Publish Point
Reset
31 fps
mkh@mkh: ~
ubuntu@ubuntu: ~ 33x45
mkh@mkh: ~ 39x22
ubuntu@ubuntu:~$ ros2 launch turtlebot3_bringup robot.launch.py
[INFO] [launch]: All log files can be found below /home/ubuntu/.ros/log/2024-01-31-08-58-53-899293-ubuntu-1632
[INFO] [launch]: Default logging verbosity is set to INFO
urdf_file_name : turtlebot3_burger.urdf
[INFO] [robot_state_publisher-1]: process started with pid [1633]
[INFO] [hlds_laser_publisher-2]: process started with pid [1635]
[INFO] [turtlebot3_ros-3]: process started with pid [1637]
[turtlebot3_ros-3] [INFO] [1706691534.511698629] [turtlebot3_node]: Init TurtleBot3 Node Main
[turtlebot3_ros-3] [INFO] [1706691534.512243219] [turtlebot3_node]: Init DynamixelSDKWrapper
[hlds_laser_publisher-2] [INFO] [1706691534.512605531] [hlds_laser_publisher]: Init hlds_laser_publisher Node Main
[hlds_laser_publisher-2] [INFO] [1706691534.513217454] [hlds_laser_publisher]: port : /dev/ttyUSB0 frame_id : base_scan
[turtlebot3_ros-3] [INFO] [1706691534.515786957] [DynamixelSDKWrapper]: Succeeded to open the port(/dev/ttyACM0)!
[turtlebot3_ros-3] [INFO] [1706691534.523483005] [DynamixelSDKWrapper]: Succeeded to change the baudrate!
[robot_state_publisher-1] [INFO] [1706691534.554664563] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-1] [INFO] [1706691534.554976654] [robot_state_publisher]: got segment base_l
lator.cc:82] Dropped 1 earlier points.
[cartographer_node-1] [WARN] [1706692111.859652302] [cartographer logger]: W0131 18:08:31.000000 42500 range_data_collator.cc:82] Dropped 1 earlier points.
[cartographer_node-1] [WARN] [1706692112.258351784] [cartographer logger]: W0131 18:08:32.000000 42500 range_data_collator.cc:82] Dropped 1 earlier points.
[cartographer_node-1] [WARN] [1706692113.053764347] [cartographer logger]: W0131 18:08:33.000000 42500 range_data_collator.cc:82] Dropped 1 earlier points.
[cartographer_node-1] [WARN] [1706692114.253409127] [cartographer logger]: W0131 18:08:34.000000 42500 range_data_collator.cc:82] Dropped 1 earlier points.
[cartographer_node-1] [WARN] [1706692114.661735076] [cartographer logger]: W0131 18:08:34.000000 42500 range_data_collator.cc:82] Dropped 2 earlier points.
mkh@mkh: ~ 39x22
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)
space key, s : force stop
CTRL-C to quit
currently: linear velocity 0.01 angular velocity 0.0
currently: linear velocity 0.02 angular velocity 0.0
currently: linear velocity 0.03 angular velocity 0.0
currently: linear velocity 0.03 angular velocity 0.1
currently: linear velocity 0.0 angular velocity 0.0

맵이 완성되면

[PC] 맵 이름을 설정해서 저장하기

Code

```
ros2 run nav2_map_server map_saver_cli -f "<저장할 맵이름>"
```

```
mkh@mkh:~$ ros2 run nav2_map_server map_saver_cli -f "pinklab"
[INFO] [1723012725.201852620] [map_saver]:
        map_saver lifecycle node launched.
        Waiting on external lifecycle transitions to activate
        See https://design.ros2.org/articles/node_lifecycle.html for more
information.
[INFO] [1723012725.201922855] [map_saver]: Creating
[INFO] [1723012725.201972274] [map_saver]: Configuring
[INFO] [1723012725.202631685] [map_saver]: Saving map from 'map' topic to
'pinklab' file
[WARN] [1723012725.202646239] [map_saver]: Free threshold unspecified. Set
ting it to default value: 0.250000
[WARN] [1723012725.202653076] [map_saver]: Occupied threshold unspecified.
 Setting it to default value: 0.650000
[WARN] [map_io]: Image format unspecified. Setting it to: pgm
[INFO] [map_io]: Received a 140 X 130 map @ 0.05 m/pix
[INFO] [map_io]: Writing map occupancy data to pinklab.pgm
[INFO] [map_io]: Writing map metadata to pinklab.yaml
[INFO] [map_io]: Map saved
[INFO] [1723012725.240085565] [map_saver]: Map saved successfully
[INFO] [1723012725.241281452] [map_saver]: Destroying
mkh@mkh:~$
```

[PC] 맵이 잘 저장됐는지 확인

PinkLAB YouTube Channel

PinkLAB Studio — Robotics, AI, ROS2 and more!

Subscribe on YouTube

https://www.youtube.com/@pinklab_studio

© PinkLAB · Exported with Confluence PDF Exporter